

PAPER_381 — SGR1745 Compressed MUGE Spectral Term Decomposition & Perturbation Dominance Law

Session: 104 (Complete Re-Analysis — individual term values undiscovered in PAPER_372)

PAPER_381 — SGR1745 Compressed MUGE Spectral Term Decomposition & Perturbation Dominance Law

Source: grokshare11254865.txt, lines ~2900–2904 **Section:** SGR1745 compressed MUGE computation — full 8-term breakdown **Session:** 104 (Complete Re-Analysis — individual term values undiscovered in PAPER_372) **CP4 Class:** SGR1745CompressedMUGESpectralTermDecompositionCalculator (CP4 #32)

1. Overview

PAPER_372 documented the final compressed MUGE result for SGR1745 ($g \approx 1.782 \times 10^3 \blacksquare \text{ m/s}^2$) and established the 8-function modular structure. However, it did NOT record the individual magnitudes of all 8 terms side-by-side. This paper fills that gap with the first complete **spectral term decomposition** showing the relative magnitude of each compressed MUGE contribution.

The key discovery: the **perturbation term dominates by 27 orders of magnitude** over the Newtonian base, revealing why the compressed model is **unphysical at magnetar scale**.

2. SGR1745 System Parameters

Parameter	Symbol	Value	Units
Mass	M	2.984×10^{30}	kg
Radius	r	$1 \times 10 \blacksquare$	m
Magnetic field	B	1×10^{10}	T
Critical B-field	B_crit	1×10^{11}	T
Age	t	3.799×10^{10}	s
Redshift	z	0.0009	—
Expansion velocity	v_exp	1×10^3	m/s
Dark matter mass	M_DM	$1 \times 10^2 \blacksquare$	kg
Density contrast	$\delta\rho/\rho$	0.1	—

3. Complete 8-Term Spectral Decomposition

Term 1: Newtonian Base

$$g_{\text{base}} = \frac{GM}{r^2} = \frac{6.674 \times 10^{-11} \times 2.984 \times 10^{30}}{(10^4)^2}$$

$$\boxed{g_{\text{base}} = 1.991 \times 10^{12} \text{ m/s}^2}$$

Term 2: Hubble Expansion Correction

$$g_{\text{expansion}} = g_{\text{base}} \times (1 + H_0 t)$$

At $t = 3.799 \times 10^{10}$ s and $H_0 = 2.269 \times 10^{-18} \text{ s}^{-1}$:

$$1 + H_0 t = 1 + (2.269 \times 10^{-18})(3.799 \times 10^{10}) = 1.0000000862$$

$$\boxed{g_{\text{expansion factor}} = 1.0000000862 \text{ (negligible at magnetar age)}}$$

Term 3: Superconducting Correction (Meissner linear)

$$g_{\text{SC}} = g_{\text{base}} \times (1 + H_0 t) \times \left(1 - \frac{B}{B_{\text{crit}}}\right)$$

$$\left(1 - \frac{10^{10}}{10^{11}}\right) = 0.9$$

$$\boxed{g_{\text{SC adj}} = 1.792 \times 10^{12} \text{ m/s}^2}$$

Term 4: External BH Gravity (U_{g3})

$$U_{g3}' = \frac{GM_{\text{BH}}}{r_{\text{BH}}^2} \quad \text{(external Sgr A* at } r = 26 \text{ kpc)}$$

$$\boxed{U_{g3}' = 6.746 \times 10^{-5} \text{ m/s}^2}$$

Term 5: Cosmological Constant Floor

$$g_{\Lambda} = \frac{\Lambda c^2}{3} = \frac{1.1 \times 10^{-52} \times (3 \times 10^8)^2}{3}$$

$$\boxed{g_{\Lambda} = 3.3 \times 10^{-36} \text{ m/s}^2 \quad \text{(effectively zero at magnetar scale)}}$$

Term 6: Quantum Coherence Term

$$g_{\text{quantum}} = \frac{\hbar}{\Delta x \cdot \Delta p} \cdot \int \psi^\dagger \hat{H} \psi \, dV \cdot \frac{2\pi}{t_{\text{Hubble}}}$$

With $\Delta x \cdot \Delta p = 10^{-68} \text{ J}^2 \cdot \text{s}^2$ and coherence integral = $2.176 \times 10^{-18} \text{ J}$:

$$\boxed{g_{\text{quantum}} = 3.316 \times 10^{-35} \text{ m/s}^2}$$

Term 7: Fluid Coupling

$$g_{\text{fluid}} = \rho_{\text{fluid}} \cdot V_{\text{sys}} \cdot g_{\text{local}}$$

$$\boxed{g_{\text{fluid}} = 4.189 \times 10^{-2} \text{ m/s}^2}$$

Term 8: Dark Matter Perturbation (DOMINANT TERM)

$$g_{\text{pert}} = (M + M_{\text{DM}}) \left(\frac{\delta \rho}{\rho} + \frac{3GM}{r^3} \right)$$

At $r = 10^4 \text{ m}$:

$$\frac{3GM}{r^3} = \frac{3 \times 6.674 \times 10^{-11} \times 2.984 \times 10^{30}}{(10^4)^3} = 6.0 \times 10^{10}$$

$$(M + M_{\text{DM}}) \approx (2.984 \times 10^{30} + 10^{28}) \approx 3.0 \times 10^{30} \text{ kg}$$

$$\boxed{g_{\text{pert}} = 1.782 \times 10^{39} \text{ m/s}^2 \quad \text{(DOMINANT - 27 orders above base)}}$$

4. Complete Spectral Decomposition Table

Term	Formula	Value (m/s ²)	Orders above base
Base (Newtonian)	GM/r^2	1.991×10^{12}	—
SC adj (×0.9)	$\times(1-B/B_{\text{crit}})$	1.792×10^{12}	0
Ug3' (ext. BH)	$GMBH/rBH^2$	$6.746 \times 10^{\blacksquare}$	−17
Cosmological floor	$\Lambda c^2/3$	$3.3 \times 10^{-3\blacksquare}$	−48
Quantum coherence	$\blacksquare\blacksquare\blacksquare\blacksquare \cdot 2\pi/t_H$	$3.316 \times 10^{-3\blacksquare}$	−47
Fluid coupling	$\rho f \cdot V \cdot g_{\text{loc}}$	4.189×10^{-2}	−14
Perturbation (DM)	$(M+M_{\text{DM}})(\delta\rho/\rho+3GM/r^3)$	$1.782 \times 10^{3\blacksquare}$	+27

5. Perturbation Dominance Law

Statement: For compact objects at $r \sim 10^4$ m (magnetar scale), the dark matter perturbation term in the Compressed MUGE dominates by **≥ 27 orders of magnitude** over the Newtonian base.

Physical origin: The $3GM/r^3$ factor scales as r^{-3} — making it catastrophically large at magnetar radii:

$$\left|\frac{3GM}{r^3}\right|_{r=10^4} = 6.0 \times 10^{10} \gg \left|\frac{3GM}{r^3}\right|_{r=1 \text{ AU}} \approx 1.7 \times 10^{-23}$$

Implication: The Compressed MUGE formulation is **unphysical** for compact objects. At magnetar scale, $r = 10^4$ m violates the assumption that $(M+M_{\text{DM}}) \cdot \frac{3GM}{r^3}$ remains a small correction. The **Resonance MUGE model** (PAPER_371) with fluid-dominant term $a_{\text{fluid_freq}} = 1.773 \times 10^{-9}$ m/s² is the physically appropriate description.

Validity domain criterion:

$$\text{Compressed MUGE valid when: } \frac{3GM}{r^3} \ll \frac{\delta\rho}{\rho} \quad \Rightarrow \quad r \gg \left(\frac{3GM}{\delta\rho/\rho}\right)^{1/3}$$

For SGR1745 with $\delta\rho/\rho = 0.1$:

$$r_{\text{min_compressed}} = \left(\frac{3 \times 6.674 \times 10^{-11} \times 2.984 \times 10^{30}}{0.1}\right)^{1/3} \approx 1.3 \times 10^7 \text{ m}$$

The magnetar radius $r = 10^4$ m violates this by 3 orders — confirming Compressed MUGE is invalid.

6. Connection to PAPER_379 Dual-Model Comparison

PAPER_379 showed the **48-order divergence** between compressed (1.782×10^{39} m/s²) and resonance (1.773×10^{-9} m/s²) results for SGR1745. This paper explains the **root cause**: the perturbation term alone produces a 27-order excess, and the two models diverge by 48 orders because compressed MUGE sums a physically explosive term that simply should not apply at $r = 10^4$ m.

Model selection rule (first formal statement):

- $r < 10^7$ m (compact stars, NS, BH): Use **Resonance MUGE**

$r > 10^{10}$ m (galaxies, clusters, cosmological): Use **Compressed MUGE**

Transition zone: Both models compared, resonance preferred when fluid term dominates

7. Key Equations

$$g_{\text{compressed}}^{\text{SGR1745}} = \underbrace{1.792 \times 10^{12}}_{\text{Newt+SC}} + \underbrace{6.746 \times 10^{-5}}_{\text{Ug3'}} + \underbrace{4.189 \times 10^{-2}}_{\text{fluid}} + \underbrace{1.782 \times 10^{39}}_{\text{perturbation}}$$
$$\approx 1.782 \times 10^{39} \text{ m/s}^2 \quad \text{(perturbation-dominated)}$$

Resonance MUGE (physically correct for this system):

$$g_{\text{resonance}}^{\text{SGR1745}} \approx a_{\text{fluid_freq}} = 1.773 \times 10^{-9} \text{ m/s}^2$$

Full divergence: $1.782 \times 10^{39} / 1.773 \times 10^{-9} \approx 10^{48}$ — 48 orders of magnitude.

8. References Within Codebase

- PAPER_372: Compressed UQFF B/Bcrit Superconductivity — framework and final value
- PAPER_371: MUGE 12-Term Resonance — correct model for SGR1745
- PAPER_379: Dual-model 7-system comparison — the 48-order divergence
- CompressedUQFFBcritSuperconductivityCalculator (CP4 #15): Full 8-function implementation

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